



Digital Signal Processing Laboratory

Experiment 8

Matched Filtering, ISI and Eye Diagrams

Name: Ayan Roy
Date: 14/09/2026

Roll No.: 34900324047
Software: Python / Google Colab

Simulation focus

BPSK → RRC pulse shaping → AWGN → RRC matched filter → delay compensation → sampling → decision

1. Objectives, tasks and required visualizations

Objectives

- Demonstrate matched-filter reception.
- Use eye diagrams to identify noise, ISI and timing sensitivity.

Implementation tasks

- 28.BPSK through RRC pulse shaping + AWGN.
- 29.Matched RRC filtering + delay compensation.
- 30.Vary SNR, roll-off, span and sampling offset.
- 31.Introduce a multipath ISI channel.

Required visualizations

Transmit/matched-filter output • Eye diagrams • Eye height vs SNR • BER vs sampling offset

Mandatory validation

Confirm total cascade delay and show that detected symbols are taken only after delay removal.

2. Theory: BPSK, RRC and matched filtering

- BPSK maps bit 1 to +1 and bit 0 to -1.
- RRC pulse shaping controls bandwidth and provides a zero-ISI design when paired with a matched RRC filter.
- AWGN introduces random amplitude/phase-free noise in this real-valued BPSK model.
- The receiver uses a matched filter, maximizing output SNR at the symbol decision instant.
- Eye diagrams visualize vertical noise margin, horizontal timing margin and ISI-induced closure.

Decision rule

$\hat{y}[k] = 1$ if $y[nk] \geq 0$; otherwise $\hat{y}[k] = 0$.

Overall pulse

RRC at Tx $\times$ RRC at Rx $\rightarrow$ Raised-Cosine response.

3. Core equations and delay calculation

BPSK mapping

$$a[k] = 2b[k] - 1, \quad b[k] \in \{0,1\}$$

Matched-filter output

$$y[n] = \sum r[m] h_{MF}[n - m]$$

BER

$$\text{BER} = \text{Nerror} / \text{Nbits}$$

Sampling index

$$n_k = D_{\text{total}} + k \cdot \text{sps} + \Delta$$

One RRC group delay

$$D_{\text{filter}} = \text{span} \cdot \text{sps} / 2$$

Cascade delay

$$D_{\text{total}} = 2D_{\text{filter}} = \text{span} \cdot \text{sps}$$

Example used in this PPT

$\text{span} = 8 \text{ symbols}, \text{sps} = 8 \text{ samples/symbol} \rightarrow D_{\text{filter}} = 32 \text{ samples} \rightarrow D_{\text{total}} = 64 \text{ samples}.$

4. BPSK matched-filter receiver block diagram



Delay checkpoint

Tx RRC delay = 32 samples; Rx matched RRC delay = 32 samples; total = 64 samples. The first valid decision is taken after this 64-sample delay.

5. Python implementation: the essential receiver logic

```
symbols = 2*bits.astype(float) - 1
up = np.zeros(len(symbols)*sps)
up[::sps] = symbols

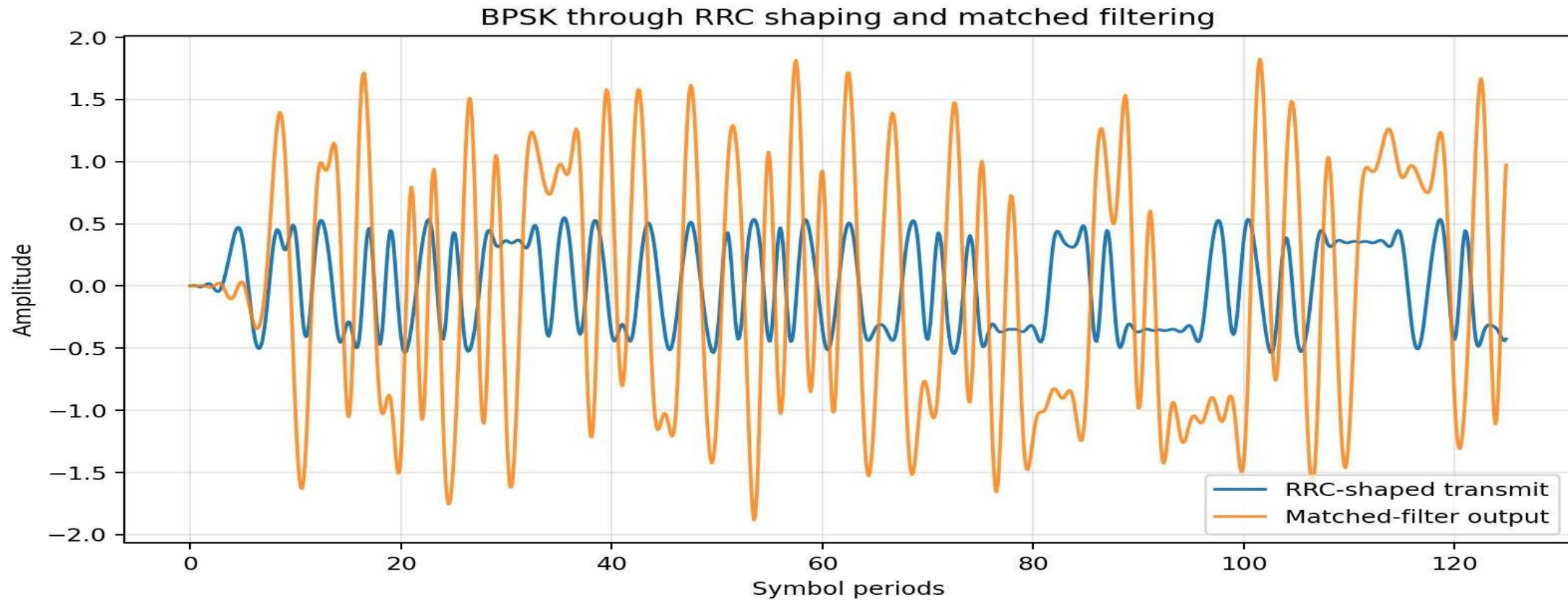
h = rrc_filter(beta, span, sps)
tx = np.convolve(up, h)
rx = add_awgn(tx, snr_db, rng)
mf = np.convolve(rx, h)

D_total = span*sps
sample_index = D_total + np.arange(len(symbols))*sps
detected = (mf[sample_index] >= 0).astype(int)
ber = np.mean(detected != bits[:len(detected)])
```

Reusable parameters

β , span , sps , snr_db and sampling offset Δ are exposed as parameters, so the same functions can be reused for the complete experiment.

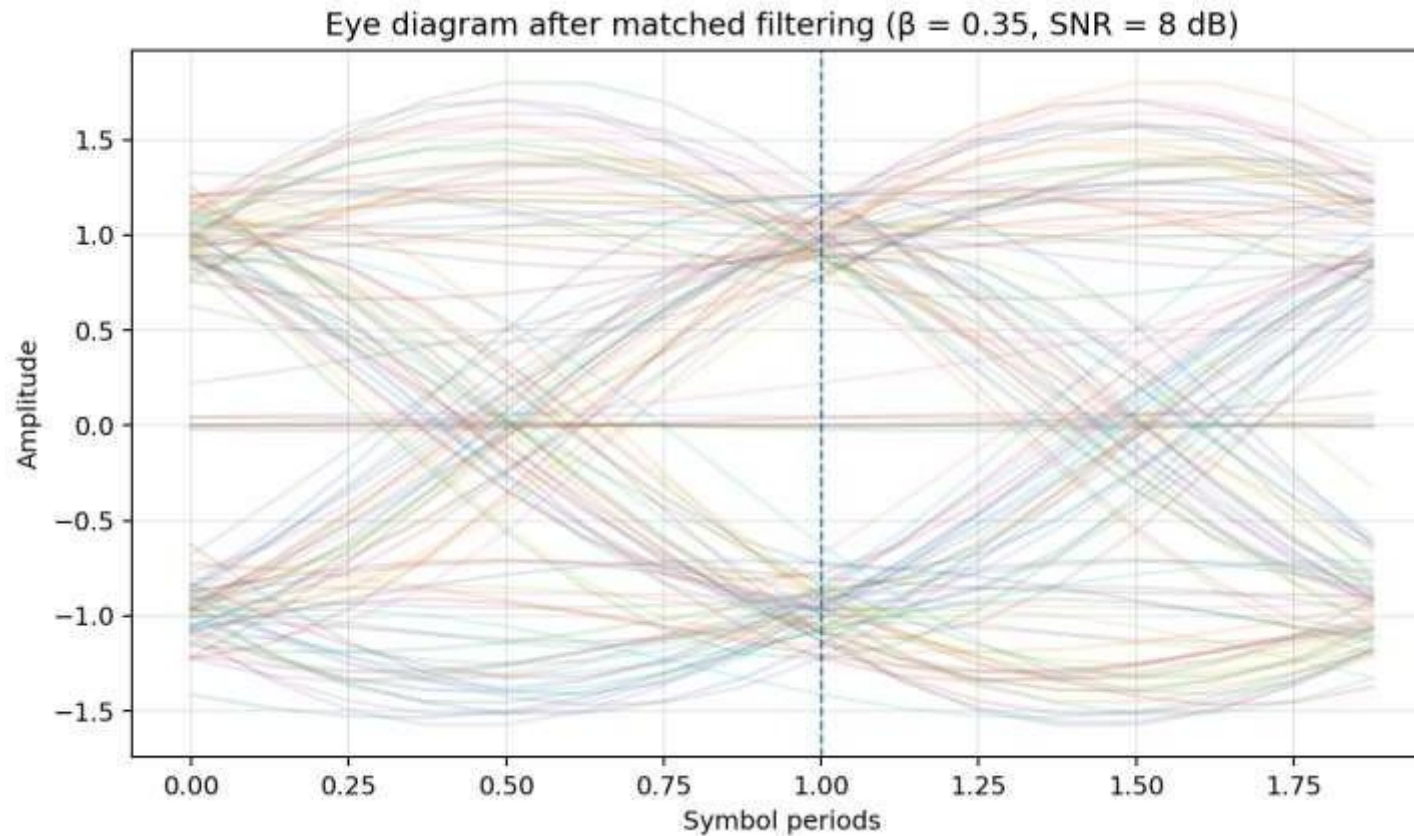
6. Transmit waveform and matched-filter output



Observation

The matched-filter output concentrates pulse energy at the correct symbol instants, improving the reliability of threshold decisions.

7. Eye diagram after matched filtering

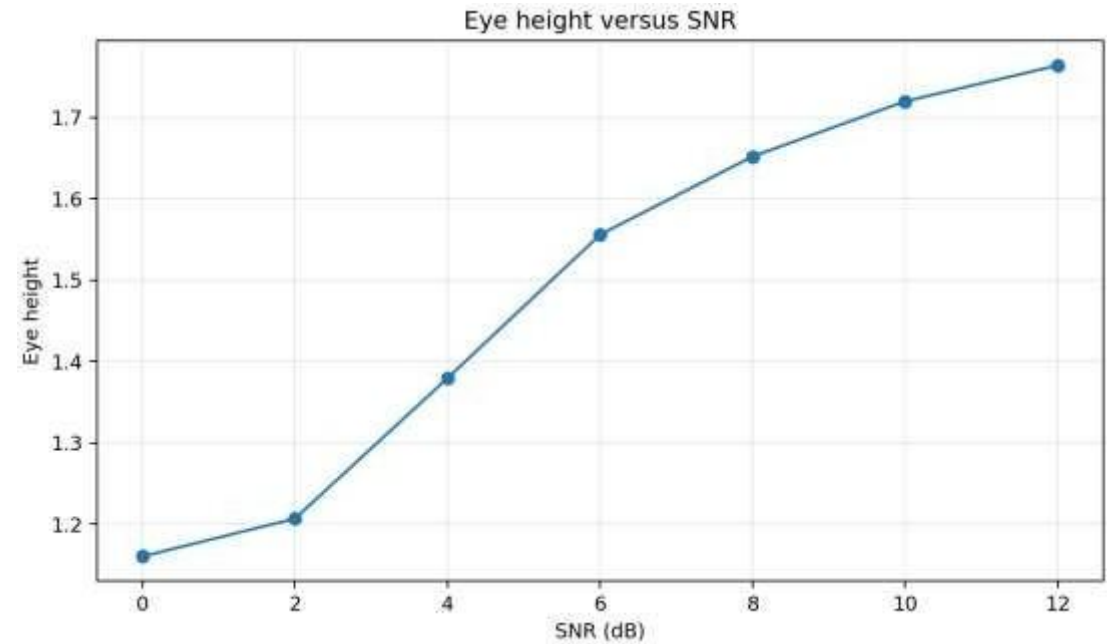
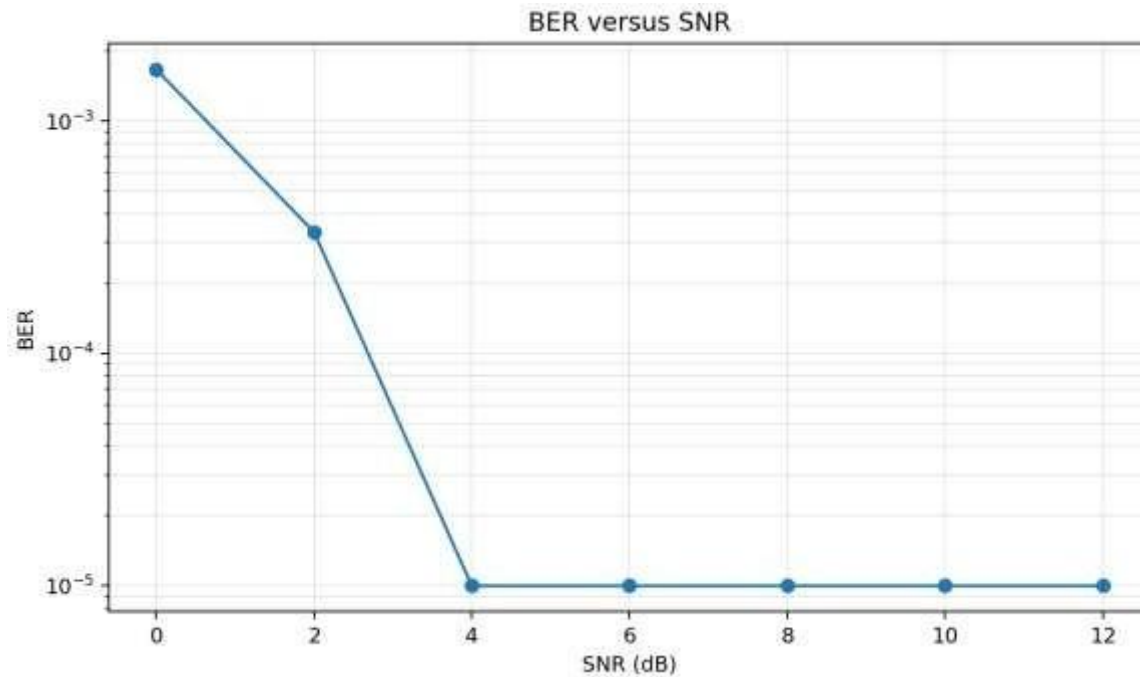


- Vertical opening $\rightarrow$ noise margin.
- Horizontal opening $\rightarrow$ timing tolerance.
- Eye closure $\rightarrow$ stronger noise, ISI or timing error.
- Center dashed line $\rightarrow$ nominal symbol sampling phase.

AI-ready prompt connection

The eye function should operate on the matched-filter output and use the compensated delay when selecting symbol-centered traces.

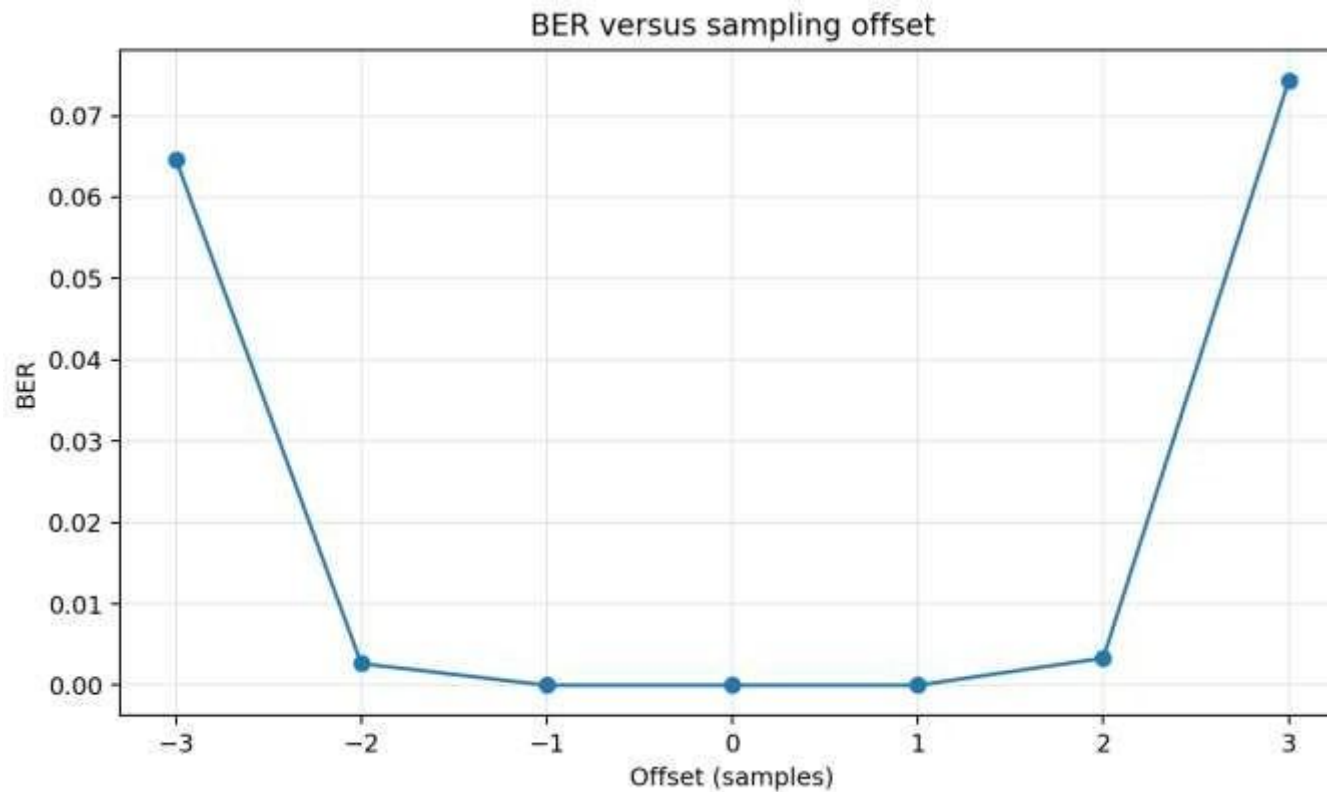
8. Performance: BER and eye height versus SNR



Physical interpretation

Higher SNR reduces random decision errors and generally increases the visible eye opening. The exact numerical curve depends on the finite simulation length and random seed.

9. Timing sensitivity: BER versus sampling offset



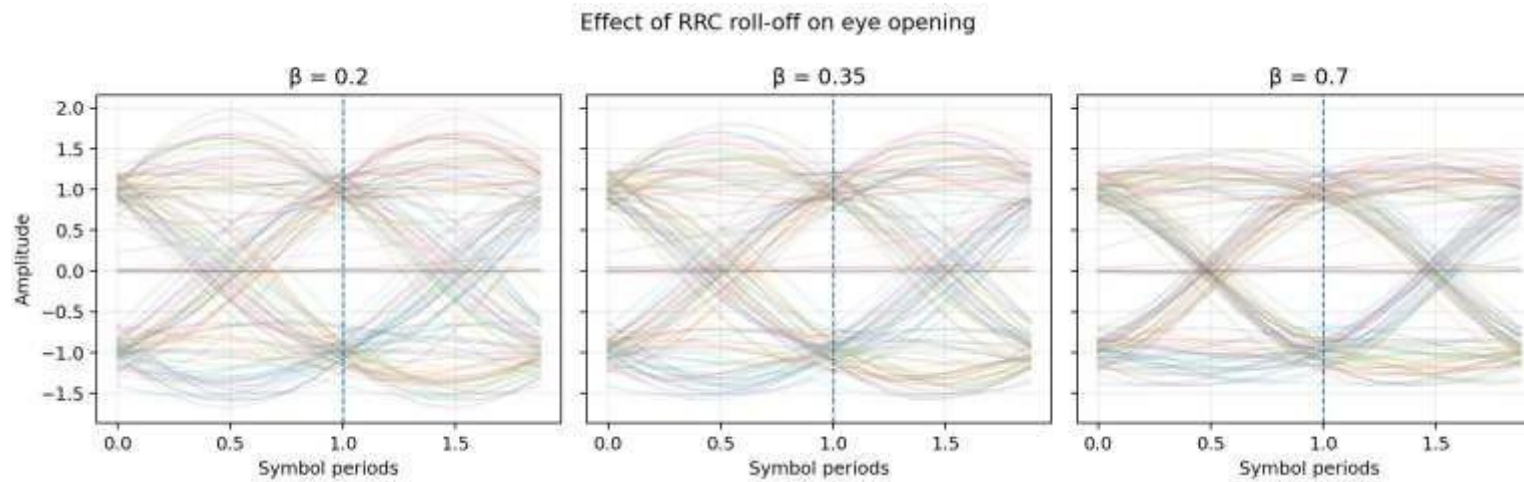
Expected trend

BER is normally lowest near the optimum eye-center sampling phase. Moving the sample by $\pm 1, \pm 2, \dots$ samples reduces timing margin.

Diagnostic test

Compare $\Delta = 0$ against positive and negative offsets at the same SNR. A sharp BER increase indicates timing sensitivity rather than a change in channel noise.

10. Roll-off factor β : pulse shape and eye diagram



- β determines excess bandwidth.
- Changing β changes the time-domain pulse shape and transition behavior.
- Compare eye opening at equal SNR, span, sps and sampling phase.
- Use the same matched-filter design at the transmitter and receiver.

11. Multipath ISI extension

Channel model

$$r[n] = s[n] + \alpha s[n-d] + w[n]$$

α = echo amplitude, d = echo delay.

Effect on eye

A delayed copy of a previous symbol overlaps the current pulse. The resulting symbol-dependent distortion can close the eye even when AWGN is unchanged.

Suggested experiment

Fix SNR and sweep $\alpha = 0, 0.2, 0.4, 0.6$. For each value, compare eye diagrams and BER. This separates multipath ISI from noise effects.

Diagnosis

Noise $\rightarrow$ vertical spread. Timing error $\rightarrow$ horizontal closure. Multipath $\rightarrow$ deterministic, symbol-dependent distortion and eye closure.

12. Links :

- **Google Colab Link:**

<https://colab.research.google.com/drive/1ywo6MRyQjVJBRhE6ISn0M85H1KPJIR6O?usp=sharing>

- **Github link:**

<https://github.com/royayan959/Matched-Filtering-ISI-and-Eye-Diagrams>

15. Reproducibility and result record

Experiment 8 simulation summary

RRC roll-off beta = 0.35

Span = 8 symbols

Samples/symbol = 8

Each RRC filter group delay = $\text{span} \times \text{sps} / 2 = 32$ samples

Total cascade delay = $\text{span} \times \text{sps} = 64$ samples

Detection uses samples only after total delay compensation.

BER vs SNR:

0 dB : 0.00166667

2 dB : 0.000333333

4 dB : 0

6 dB : 0

8 dB : 0

10 dB : 0

12 dB : 0

BER vs sampling offset at 6 dB:

Reproducibility

The script uses fixed random seeds for the demonstration and sweeps, so the included plots are reproducible on the same Python/NumPy/SciPy environment.

13. Observation and interpretation

- **SNR:** Higher SNR produces a clearer eye diagram and reduces BER.
- **Timing Offset:** Incorrect sampling timing reduces eye opening and increases detection errors.
- **Matched Filtering:** Improves signal quality and detection reliability.
- **ISI:** Multipath causes waveform distortion and eye closure.
- **Delay Validation:** Total filter delay is compensated before symbol detection to ensure accurate sampling.

Conclusion

Eye diagrams provide an intuitive view of noise, ISI and timing sensitivity, while BER provides a quantitative performance measure.